

Project report

QUIZ MASTER APPLICATION-MAD1

Introduction

As a Bachelor of Science student at IIT Madras, I am pleased to present this project report detailing my work on the MAD-1 project. While pursuing this online degree program, I am simultaneously advancing my academic journey through an offline degree program, demonstrating my commitment to comprehensive education in technology.

Throughout this report, I will detail the development and implementation of the MAD-1 project, which serves as a practical application of my technical expertise. This project reflects not only my technical abilities but also my capacity to manage complex tasks while maintaining high academic standards across both my educational pursuits.

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- GitHub - https://github.com/23f1003160/quiz_master_23f1003160

Project Description

The Quiz Master project (V1) is a comprehensive multi-user application designed to serve as an exam preparation platform for multiple courses. At its core, the application facilitates interaction between two main roles: an administrator (quiz master) with root access and regular users who can attempt quizzes.

The application requirements specified the use of Flask for backend development, Jinja2 templating with HTML/CSS and Bootstrap for frontend, and SQLite as the database system. All functionality was to be implemented for local machine operation.

Key features of the project include:

- A robust user management system with registration and authentication
- Subject and chapter organization capabilities for structured content
- Quiz creation and management tools for administrators
- Quiz attempt tracking and score management for users
- Dashboard interfaces for both admin and regular users

My approach to implementing this project focused on creating a clean, user-friendly interface while ensuring all core functionalities were properly implemented. I paid special attention to the database design to ensure efficient data management and retrieval, particularly for the quiz and scoring systems.

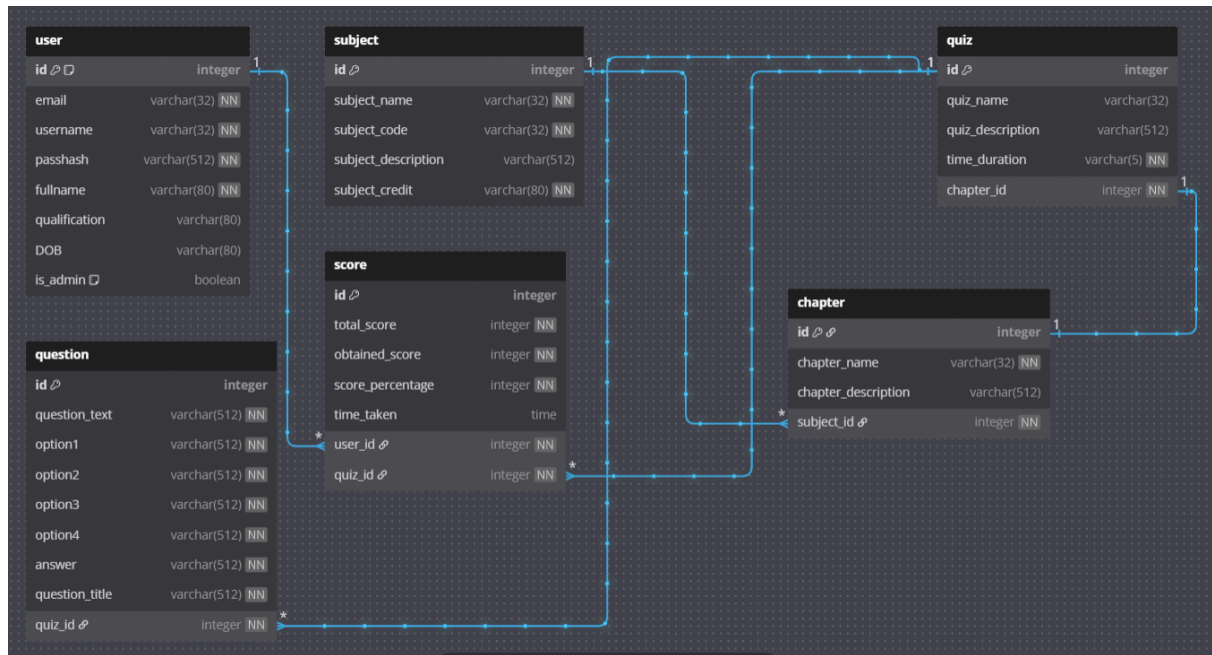
Technologies Used

The project was developed using the following frameworks and libraries:

- **Flask:** Python web framework used for backend development and routing
- **SQLite:** Lightweight, serverless database engine for data storage
- **Jinja2:** Template engine for generating dynamic HTML content
- **Bootstrap 5:** Frontend framework for responsive design and UI components
- **SQLAlchemy:** SQL toolkit and ORM for Python, used for database operations
- **Werkzeug:** WSGI web application library, used for password hashing and security
- **Flask-Login:** User session management for Flask applications

These technologies were chosen for their reliability, extensive documentation, and seamless integration capabilities, enabling the development of a robust web application.

ER Diagram



Video Presentation Link

https://drive.google.com/file/d/1xa_LVXIJBWqVMI_5dT9k-7frgxDBtN4R/view?usp=sharing